EXAM SHIELD

Distributed Systems and Blockchain Question Bank

Consistency, consensus, replication, fault tolerance and blockchains.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 21 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Distributed Systems and Blockchain

Consistency, consensus, replication, fault tolerance and blockchains.

1. By the CAP theorem, during a network partition a distributed system must choose between:

A. Latency and storage
B. Consistency and availability
C. Security and usability
D. Speed and cost

Answer: B

2. Which algorithm lets replicated servers agree on a shared log?

A. Dijkstra's algorithm
B. Huffman coding
C. RSA
D. Raft

Answer: D

3. Bitcoin miners compete to add blocks using:

A. Proof of stake
B. Proof of work
C. Round-robin turns
D. Two-phase commit

Answer: B

4. A smart contract is:

A. An encrypted email
B. A legal document signed online
C. Code stored and executed on a blockchain
D. A cloud service-level agreement

Answer: C

5. Replicating data across several nodes mainly improves:

A. Compile speed
B. Availability and fault tolerance
C. Screen resolution
D. Code readability

Answer: B

6. Two-phase commit is used to:

A. Make a transaction atomic across several nodes
B. Collect garbage memory
C. Hash passwords
D. Route network packets

Answer: A

7. Eventual consistency means that:

A. Concurrent writes are rejected
B. Data is never replicated
C. All replicas converge if no new updates arrive
D. Every read returns the latest write

Answer: C

8. What makes past blocks in a blockchain tamper-evident?

A. Each block contains the hash of the previous block
B. Blocks are compressed
C. Blocks are numbered in order
D. Blocks are kept in RAM

Answer: A

9. A system that keeps working correctly when some of its components fail is fault _____. [1 mark]

Answer: tolerant

10. In Raft, one server is elected _____ and the others act as followers. [1 mark]

Answer: leader

11. Splitting a large database across machines by key ranges or hashes is called _____. [1 mark]

Answer: sharding / partitioning / horizontal partitioning

12. Explain why clocks cannot be perfectly synchronised in a distributed system, and how logical clocks order events. [5 marks]

13. Evaluate whether a blockchain suits storing university certificates, compared with a conventional database. [5 marks]